

LPST-LST Compatibility Layer: A Formal Interface Between Phase-Primary and Luminous-Phenomenological Descriptions

Research Team

Dust LLC

May 2026

[arXiv:2605.XXXXX [hep-th]]

Canonical compatibility specification

PACS: 04.20.Cv, 11.10.Lm, 12.10.-g, 03.50.De

Abstract

We establish a formal compatibility layer between Light-Phase Substrate Theory (LPST) and Luminous Substrate Theory (LST), two complementary descriptions of a single-substrate ontology for emergent spacetime, matter, and gravity. LPST operates at the ontological hardware level, positing a universal complex field $L = A e^{i\varphi}$ with internal polarization $\mathbf{u}$, from which all physics emerges via phase-consistency constraints, topological defect sectors, and correlation-derived geometry. LST operates at the effective phenomenological level, positing the electromagnetic field — light — as the single ontological primitive from which mass, time, space, and gravity emerge as behaviors of a self-interacting luminous substrate. Both frameworks share ontological monism, background independence, the identification of matter with topological defects, emergent gravity, and the aspiration to derive all constants from substrate parameters. The critical difference lies in ontological depth: LPST treats phase geometry as primary, while LST treats the electromagnetic field itself as the substrate. This compatibility layer prevents category errors, false contradictions, and misleading intuitions when deploying both frameworks jointly. We establish a rigorous concept-mapping dictionary translating every LST construct into its LPST antecedent, define three classes of permitted inter-theory statements (emergent language, correlation language, and effective limits), enumerate a canonical list of prohibited phrasings with their corrected forms, formulate cross-theory paradox-handling rules, identify five irreducible

shared paradoxes constituting the closure boundaries of the single-substrate program, and state the precise claims each theory is forbidden from making. The result is a complete interface specification ensuring that LPST and LST can be deployed together without ontological confusion: LPST defines what reality is allowed to be; LST describes how that reality appears when phase dynamics are dominated by luminous propagation.

Keywords: unified field theory, emergent spacetime, phase substrate, luminous substrate, effective field theory, ontological layering, compatibility specification

1. Introduction and Motivation

The single-substrate research program aims to derive all of physics — spacetime geometry, matter content, gravitational dynamics, and quantum correlations — from a single ontological primitive. Two complementary frameworks have been developed within this program: Light-Phase Substrate Theory (LPST) [1] and Luminous Substrate Theory (LST) [2]. Both share the conviction that nature is fundamentally monistic and that the apparent diversity of physical phenomena arises from the internal dynamics of one universal substance. They differ, however, in the level of ontological description at which they operate, and it is this difference that necessitates the present compatibility specification.

LPST, as formulated in Ref. [1], is defined by four postulates that constitute the deepest layer of the single-substrate ontology:

(P1) The Substrate Postulate. The sole ontological primitive is a universal complex scalar field with internal polarization,

$$L(x) = A(x) e^{i\varphi(x)}, \quad \mathbf{u}(x) \in M_{int}, \quad (1)$$

where $A(x) \geq 0$ is the amplitude, $\varphi(x) \in \mathbf{R}$ is the phase, and $\mathbf{u}(x)$ is an internal polarization vector taking values in a compact manifold M_{int} . Every physical degree of freedom — every particle, field, and geometric structure — is a configuration of L .

(P2) Background Independence. No background spacetime is postulated. Space and time are not containers in which the field L resides; they are emergent structures derived from correlations among the values of L at different points. The topological base space on which L is initially defined is a bare set endowed only with a neighborhood structure sufficient to define continuity; all geometric content (metric, connection, curvature) is constructed from the field itself.

(P3) The Phase Consistency Principle. The dynamics of the substrate are not governed by a conventional Hamiltonian or Lagrangian principle but by a global interference-closure condition. The **phase inconsistency functional**,

$$I[\varphi] = \oint (\nabla\varphi) \cdot d\mathbf{l} - 2\pi n, \quad (2)$$

must vanish on every contractible loop (with $n = 0$ for contractible loops and $n \in \mathbf{Z}$ for non-contractible loops encircling topological defects). Physical configurations are those that globally minimize the total inconsistency. This principle replaces Newton's second law, the Euler-Lagrange equations, and the Schrödinger equation as the fundamental dynamical selector.

(P4) Geometry from Correlations. The emergent metric tensor is derived from the two-point correlation kernel,

$$C(x,y) = \langle L(x)L^*(y) \rangle, \quad (3)$$

with the emergent metric $g_{\mu\nu}$ constructed from second derivatives of $\ln|C|$ evaluated at coincidence (LPST Eq. 24 of Ref. [1]). Distances, angles, volumes, and causal structure are all derived quantities. The correlation kernel encodes the full geometric content of the emergent spacetime.

LST, as formulated in Ref. [2], takes a different starting point. It is defined by three axioms:

(A1) The Luminous Substrate. The sole ontological primitive is a substrate tensor field $S_{\mu\nu}$ propagating at the invariant speed c . This field is identified with the electromagnetic field in the broadest sense: it encompasses both the familiar antisymmetric Faraday tensor and a symmetric component encoding gravitational degrees of freedom.

(A2) Emergent Metric. The spacetime metric $g_{\mu\nu}$ is a quadratic functional of $S_{\mu\nu}$, constructed from contractions and averages of the substrate field. Geometry is not a background but an output of the luminous substrate’s self-organization.

(A3) Nonlinear Self-Interaction. The substrate supports both radiative (wave-like, propagating) and solitonic (topologically stabilized, localized) solutions through its nonlinear self-interaction. Particles are solitons of the luminous substrate — “trapped light” in the evocative language of LST.

The two frameworks share a remarkable convergence of structural commitments. Both are ontologically monistic: a single substance underlies all of physics. Both are background-independent: spacetime geometry is derived, not assumed. Both identify matter with topological defects or solitonic structures of the fundamental field. Both treat gravity as emergent from substrate dynamics rather than as a fundamental force. And both aspire to derive all physical constants — particle masses, coupling strengths, the cosmological constant — from the parameters of the substrate.

The critical difference between LPST and LST lies in their respective levels of ontological depth. LPST treats **phase geometry** as ontologically primary. The substrate is the complex field $L = A e^{i\varphi}$ with its internal polarization $\mathbf{u}$; light — the electromagnetic field, propagating radiation — is a *derived excitation* of this substrate. It is a coherent phase-propagation mode, a small

perturbation $\delta\varphi$ propagating on the globally coherent vacuum state $L_{\text{vac}} = A_0 e^{i\varphi_0}$. The substrate is not light; it is the phase-geometric structure from which light emerges.

LST, by contrast, treats the **electromagnetic field itself** as ontologically primary. The substrate IS light, in the broadest possible sense of that term. The electromagnetic field is not an excitation of something more fundamental; it is the fundamental thing. Mass, time, space, and gravity are emergent behaviors of this self-interacting luminous medium.

This difference in ontological depth creates the need for a formal compatibility layer. The relationship between LPST and LST is analogous to three well-understood relationships in physics:

Microphysics and hydrodynamics. LPST is to LST as molecular dynamics is to the Navier-Stokes equations. The Navier-Stokes description of fluid flow is valid, predictive, and extraordinarily useful, but the variables it treats as primitive — density, pressure, velocity, viscosity — are coarse-grained observables derived from the underlying molecular dynamics. No one mistakes the Navier-Stokes equations for a microscopic theory. Similarly, LST's variables (refractive index, trapped waves, substrate oscillation cycles) are coarse-grained observables derived from LPST's phase dynamics.

Hardware and software. LPST defines the instruction set and physical gates of reality; LST is the operating system running on that hardware. The operating system provides a useful, intuitive description of what the user experiences, but the hardware defines what is physically possible. Software cannot override hardware constraints.

Field equations and effective media. LPST is to LST as Maxwell's equations in a material are to the constitutive relations (permittivity ϵ , permeability μ). The refractive index $n(x)$ that plays a central role in LST's description of gravity is an effective-medium parameter, analogous to the refractive index of

glass. It is derived from LPST's phase stiffness $\kappa(x) = \alpha A(x)^2$, not posited as fundamental.

Without this compatibility layer, practitioners of the single-substrate program risk three categories of error. First, **category errors**: treating LST effective variables (refractive index, trapped waves, substrate oscillation counts) as if they were fundamental quantities, when in fact they are derived observables. Second, **false contradictions**: perceiving apparent conflicts between LPST and LST that arise not from genuine physical disagreement but from confusing the ontological layers at which the two theories operate. Third, **misleading intuitions**: importing intuitions from LST's effective description ("light is the substrate," "matter is trapped light") into contexts where they are technically incorrect and potentially confusing.

The present paper establishes the formal interface between LPST and LST. The structure is as follows. Section 2 formulates the Layering Principle — the strict, unidirectional derivation relationship between the two theories. Section 3 presents a comprehensive concept-mapping dictionary translating every major LST construct into its LPST antecedent. Section 4 defines three classes of permitted inter-theory statements. Section 5 enumerates the canonical list of prohibited phrasings with their corrected forms. Section 6 formulates the cross-theory paradox-handling rules. Section 7 identifies the five irreducible shared paradoxes that constitute the closure boundaries of the single-substrate program. Section 8 states the precise claims each theory is forbidden from making. Section 9 presents the formal compatibility statement. Section 10 discusses open problems and outlook. Section 11 concludes.

2. The Layering Principle

2.1. Statement of the Principle

The relationship between LPST and LST is one of strict ontological layering. LPST occupies the **ontological hardware layer**: it defines the space of physically allowed configurations through its four postulates, the phase consistency principle, the topology of the internal manifold M_{int} , and the defect-sector classification scheme. LST occupies the **effective phenomenological layer**: it describes the macroscopic appearance of the substrate when the dynamics are dominated by propagating phase modes (the “luminous sector”) in the low-energy, long-wavelength regime.

This layering is not a matter of convenience or pedagogical preference. It is a structural feature of the theoretical framework, and it is *strict* and *unidirectional*: LST derives from LPST, never the reverse. LST cannot veto, override, or modify LPST axioms.

Rule 1 — The Non-Negotiable Layering Rule

Any statement in LST that appears fundamental must be reinterpreted as emergent from LPST. LST provides effective descriptions; LPST provides ontological foundations. The derivation direction is irreversible.

This rule is not a methodological suggestion. It is a logical consequence of the structural relationship between the two theories: LPST’s postulates are axioms; LST’s axioms are theorems derivable (in principle) from those LPST axioms under appropriate coarse-graining and sector restriction.

2.2. Three Analogies for the Layering

(a) Hardware vs. Software. Consider a digital computer. The hardware — transistors, logic gates, clock signals, memory cells — defines the space of all possible computations. The operating system provides a useful abstraction: files, windows, applications, a graphical interface. The user interacts with the software layer and finds it intuitive, productive, and (within its domain) entirely reliable. But the software description is not fundamental. Files are not physical objects; they are patterns of transistor states. A “window” is not a pane of glass; it is a software construct rendered by hardware. Most importantly, the software cannot perform any operation that the hardware prohibits. If the hardware lacks a floating-point unit, the software cannot perform hardware-accelerated floating-point arithmetic, no matter how cleverly it is coded.

In this analogy, LPST is the hardware layer. Its postulates define the instruction set of reality: what configurations of the substrate are physically allowed, what topological charges are conserved, what causal structures are permitted. LST is the operating system: it provides an intuitive, phenomenologically useful description of what the substrate “looks like” when the dominant dynamics are luminous propagation. The LST description is valid, useful, and often easier to work with than the LPST description, just as it is easier to edit a document in a word processor than to manipulate individual transistor states. But the LST description cannot override LPST constraints, just as software cannot override hardware.

(b) Microphysics vs. Hydrodynamics. The Navier-Stokes equations describe the flow of a continuous fluid characterized by density ρ , velocity $\mathbf{v}$, pressure p , and viscosity η . These equations are spectacularly successful: they describe weather systems, ocean currents, aerodynamic lift, and turbulent flow. But density, pressure, and viscosity are not microscopic primitives. They are statistical averages over the positions and momenta of

$\sim 10^{23}$ molecules. The Navier-Stokes equations are an effective theory — valid in the hydrodynamic limit (length scales much larger than the mean free path, time scales much longer than the collision time) — derived from the Boltzmann equation, which is itself derived from Hamiltonian molecular dynamics.

LPST is the molecular dynamics; LST is the Navier-Stokes description. LST's variables — the refractive index $n(x)$, the “trapped light” model of matter, the oscillation-counting definition of time — are coarse-grained observables, valid in the effective regime but not to be confused with fundamental degrees of freedom. Just as one cannot use the Navier-Stokes equations to describe individual molecular collisions, one cannot use LST to describe Planck-scale substrate dynamics.

(c) Field Equations vs. Effective Media. Maxwell's equations in a material medium are often written using constitutive relations: $\mathbf{D} = \epsilon \mathbf{E}$, $\mathbf{B} = \mu \mathbf{H}$. The permittivity ϵ and permeability μ (and hence the refractive index $n = c/(c/\sqrt{\epsilon\mu})$) are effective-medium parameters that encode the response of the material's microscopic constituents (atoms, electrons, lattice vibrations) to applied electromagnetic fields. They are enormously useful for practical optics and engineering, but they are not fundamental: they are derived from quantum electrodynamics applied to the atomic structure of the medium.

In this analogy, LPST provides the microscopic field equations; LST provides the constitutive relations. The refractive index $n(x)$ that plays a central role in LST's description of gravitational lensing, time dilation, and black holes is an effective-medium parameter derived from LPST's phase stiffness:

$$n(x) = \sqrt{(\kappa_0/\kappa(x))} = A_0/A(x), \quad (4)$$

where $\kappa(x) = \alpha A(x)^2$ is the local phase stiffness and $\kappa_0 = \alpha A_0^2$ is its vacuum value. This is analogous to Gordon's optical metric [8]: gravity appears as

refraction because the phase stiffness varies spatially, but the refractive index is a *consequence* of phase dynamics, not a cause.

2.3. Formal Derivation Pathway

The formal relationship between the two theories can be stated precisely. LST's substrate field $S_{\mu\nu}$ decomposes into an antisymmetric part $F_{\mu\nu} = S_{[\mu\nu]}$ (the electromagnetic field strength) and a symmetric part $H_{\mu\nu} = S_{(\mu\nu)}$ (encoding gravitational degrees of freedom). In LPST terms, $F_{\mu\nu}$ corresponds to the coherent phase-propagation modes — linearized excitations of φ around the vacuum state — while $H_{\mu\nu}$ corresponds to the correlation-derived emergent metric $g_{\mu\nu}$ obtained from the kernel $C(x,y)$.

The LST action (Eq. 4 of Ref. [2]) is formally an **effective action** obtained from the LPST partition function by integrating out the amplitude mode $A(x)$ (which is frozen to $A \approx A_0$ in the low-energy limit), coarse-graining the phase field φ and polarization field $\mathbf{u}$ over length scales large compared to the substrate correlation length ξ , and restricting attention to the dominant propagating sector. This is a standard effective-field-theory procedure, analogous to integrating out heavy modes in the Wilsonian renormalization group.

The layering is therefore not merely analogical; it is a precise effective-field-theory relationship. LST is the low-energy effective theory of LPST, valid when the amplitude mode is frozen, the phase and polarization fields dominate the dynamics, and the emergent metric is smooth.

3. Concept Mapping Dictionary: LPST to LST Translation

This section presents the canonical dictionary for translating LST concepts into their LPST antecedents. Every concept that appears in LST has a precise

interpretation within LPST; the dictionary makes these correspondences explicit and eliminates ambiguity.

Rule 2 — The Variable Status Rule

LST variables (e.g., refractive index, trapped waves, substrate oscillation counts) are coarse-grained observables, not primitives. They must always be understood as derived from LPST's phase-geometric degrees of freedom.

The following table provides a compact summary of the concept mapping. Extended discussion of each entry follows the table.

LST Concept	LPST Interpretation
Light as substrate	Dominant propagating excitation of the phase substrate
EM field	Coherent phase-propagation mode
Vacuum	Globally consistent, low-inconsistency phase configuration
Matter = trapped light	Topologically stabilized phase defect
Standing wave	Local minimum of the global phase-inconsistency functional
Mass	Phase circulation / internal update burden
Time	Ordered phase-update bookkeeping
Space	Interference/relational volume of phase propagation
Gravity	Phase-consistency-driven metric response
Refractive index $n(x)$	Emergent macroscopic proxy for phase strain
Entanglement	Shared topological phase structure

LST Concept	LPST Interpretation
Instant correlation	Nonlocal constraint satisfaction (not signal)
Black hole	Maximally saturated phase knot / causal blockade
Dark matter	Weakly coupled phase-topology sector

3.1. “Light as substrate” → Dominant propagating excitation of the phase substrate

In LST, the phrase “light is the substrate” is the foundational axiom (A1). In LPST, this statement must be reinterpreted. The LPST vacuum is a globally coherent phase state $L_{\text{vac}} = A_0 e^{i\varphi_0}$ with uniform polarization $\mathbf{u}_0$. Small perturbations $\delta\varphi$ about this vacuum propagate as massless waves satisfying the linearized phase field equation $\nabla^2\varphi = 0$ (LPST Eq. 16 of Ref. [1]). These propagating perturbations are what we call “light” — electromagnetic radiation. They are the dominant low-energy excitation of the substrate, the easiest mode to excite and detect, and they dominate the phenomenology at macroscopic scales. But the substrate *itself* is the phase field L , not the excitations upon it. Light is to the phase substrate as sound is to air: a propagating disturbance, not the medium.

3.2. “EM field” → Coherent phase-propagation mode

The electromagnetic field strength tensor $F_{\mu\nu}$ is identified in LPST with the curvature of the emergent U(1) connection arising from phase gradients. Specifically, defining the emergent gauge potential $A_\mu \propto \partial_\mu\varphi$, the field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (5)$$

which corresponds to LPST Eq. 35 of Ref. [1]. Maxwell’s equations emerge from the linearized phase field equation (LPST Eq. 15) in the vacuum sector,

where the amplitude is frozen at $A = A_0$ and the polarization is uniform. The full nonlinear LPST dynamics contains corrections to Maxwell's equations at high field strengths (near topological defects), but in the weak-field regime the correspondence is exact.

3.3. “Vacuum” → Globally consistent, low-inconsistency phase configuration

LST's vacuum is the equilibrium state of the electromagnetic substrate — the state of zero electromagnetic excitation. In LPST, this corresponds to the maximally coherent configuration: $L_{\text{vac}} = A_0 e^{i\varphi^0}$ with $\mathbf{u}_{\text{vac}} = \mathbf{u}_0 = \text{const}$, where the correlation kernel satisfies $C(x, y) = A_0^2$ for all x, y , all phase gradients vanish ($\nabla\varphi = 0$ everywhere), and the phase inconsistency functional $\mathcal{I}[\varphi]$ is globally minimized (equal to zero on every loop). The vacuum is not empty; it is the state of maximal phase coherence.

3.4. “Matter = trapped light” → Topologically stabilized phase defect

LST models particles as toroidal vortices — standing waves of the electromagnetic substrate trapped in self-reinforcing geometries (LST Section 8 of Ref. [2]). In LPST, these structures are identified with topological solitons of the phase φ and polarization $\mathbf{u}$ fields, classified by homotopy groups: vortices by $\pi_1(S^1)$, skyrmions by $\pi_2(S^2)$, and hopfions by $\pi_3(S^2)$, following the classification scheme developed in Refs. [6,7]. The critical upgrade from LST to LPST is the explanation of *stability*: LPST explains why these structures persist indefinitely. They carry integer-valued topological charges,

$$Q = \int f^0 d^3x \in \mathbb{Z}, \quad (6)$$

that are exactly conserved by the continuity of the phase and polarization fields (LPST Eq. 31 of Ref. [1]). No continuous deformation can change Q ; the

defect is topologically protected. LST's "trapped light" picture captures the oscillatory character of these structures but does not, by itself, guarantee their stability against perturbations. LPST provides the guarantee: topology, not wave interference, is the stabilizer.

3.5. "Standing wave" → Local minimum of the global phase-inconsistency functional

The phase consistency principle (LPST Postulate P3) selects configurations that minimize the total phase inconsistency $\mathbb{I}[\varphi]$. A topological defect corresponds to a *local* minimum of this functional: the inconsistency is not zero (phase gradients are nonvanishing in the defect core), but no continuous deformation can reduce it further without changing the topological charge. The defect sits in a topological sector separated from the vacuum sector by an infinite energy barrier in the thermodynamic limit. The "standing wave" description of LST captures the oscillatory spatial profile of the defect, but the fundamental characterization is variational: the defect is a constrained minimum of $\mathbb{I}[\varphi]$ within its topological sector.

3.6. "Mass" → Phase circulation / internal update burden

In LPST, the mass-energy of a topological defect is the total phase distortion energy locked into its configuration. The energy density includes contributions from phase gradients, polarization gradients, and the topological stabilizer:

$$E = \int d^3x [\alpha A_0^2 (\nabla\varphi)^2 + \beta A_0^2 (\nabla\mathbf{u})^2 + V(A) + K(\varphi, \mathbf{u})], \quad (7)$$

following LPST Eq. 32 of Ref. [1]. Mass is the Lorentz-invariant measure of this trapped energy. The phrase "internal update burden" refers to the fact that the substrate must continuously process the phase gradients within the defect, and this processing cost is precisely the rest mass. LST's formula $m =$

$\hbar/(rc)$ (LST Eq. 15 of Ref. [2]) is a special case applicable to the simplest toroidal defect geometry, where r is the characteristic radius of the trapped-light orbit.

3.7. “Time” → Ordered phase-update bookkeeping

LST defines time operationally as the count of substrate oscillation cycles (LST Section 11 of Ref. [2]): a clock is any system that oscillates, and time is measured by counting those oscillations. This is a valid effective definition but not the most fundamental one. In LPST, proper time is related to the local phase dynamics by

$$d\tau \propto \sqrt{\kappa(x)} |d\phi|, \quad (8)$$

where $\kappa(x) = \alpha |L(x)|^2$ is the local phase stiffness (LPST Eq. 26 of Ref. [1]). Time is not fundamentally oscillation-counting; it is the ordering imposed by the phase-update structure of the substrate. The oscillation-counting picture emerges when the local dynamics are dominated by periodic phase orbits (as in atomic clocks), but the general definition encompasses non-periodic configurations as well.

3.8. “Space” → Interference/relational volume of phase propagation

LST defines spatial distance operationally via light travel time: $d(A,B) = c \cdot \Delta t_{A \rightarrow B}$ (LST Eq. 27 of Ref. [2]). This is a radar-distance definition, valid in the effective regime. LPST provides a deeper definition: the emergent distance between two substrate points is

$$d^2(x,y) = -\ell^2 \ln(|C(x,y)|/|C(x,x)|), \quad (9)$$

where ℓ is a fundamental length scale set by the substrate parameters (LPST Eq. 19 of Ref. [1]). The full metric tensor $g_{\mu\nu}$ is constructed from the correlation kernel $C(x,y)$ via Eq. 24 of Ref. [1]. Space is relational: distances

exist only insofar as substrate correlations define them. LST's light-travel-time definition is recovered when the emergent metric is smooth and the dominant propagation mode is the massless phase wave.

3.9. “Gravity” → Phase-consistency-driven metric response

LST models gravity as refraction of the luminous substrate through a medium with spatially varying refractive index,

$$n(r) = 1/\sqrt{1 - 2GM/(rc^2)} , \quad (10)$$

(LST Eq. 34 of Ref. [2]), recovering the Schwarzschild geometry in the weak-field limit. This description is phenomenologically powerful and connects gravity to familiar optical intuition. In LPST, the mechanism is more fundamental: topological defects (matter) suppress the substrate amplitude A in their vicinity, reducing the local phase stiffness $\kappa(x) = \alpha A(x)^2$. This modifies the correlation kernel $C(x,y)$, which in turn modifies the emergent metric $g_{\mu\nu}$. The null cone tilts, geodesics curve, and the effective phenomenology is gravitational attraction. The refractive index,

$$n(x) = A_0/A(x) , \quad (11)$$

(LPST Eq. 39 of Ref. [1]), is an *output* of this process, not an input. The causal direction in LPST is: phase consistency → amplitude suppression → metric deformation → refractive behavior. LST inverts this and treats the refractive behavior as explanatory; the compatibility layer restores the correct causal ordering.

3.10. “Refractive index $n(x)$ ” → Emergent macroscopic proxy for phase strain

The refractive index is defined as $n(x) = \sqrt{(\kappa_0/\kappa(x))} = A_0/A(x)$. It is a dimensionless ratio comparing the local phase stiffness to its vacuum value.

As a derived ratio, it is useful for computation — particularly for recovering the Gordon optical metric [8] and the analogue-gravity correspondence — but it must not be mistaken for a fundamental field. It has no independent equation of motion; its dynamics are entirely determined by the amplitude field $A(x)$, which is itself determined by the global phase consistency condition.

3.11. “Entanglement” → Shared topological phase structure

LST explains quantum entanglement as structural continuity: two defect endpoints connected by a substrate filament, a “luminous bridge” maintaining phase coherence across spatial separation (LST Section 20 of Ref. [2]). In LPST, the mechanism is more precisely characterized. Entangled systems share non-local phase correlations enforced by the global phase consistency principle. The correlation kernel $C(x,y)$ encodes these correlations directly: for entangled systems at locations x and y , $C(x,y)$ retains off-diagonal structure that reflects the shared topological origin of the two subsystems. This is analogous to the ER=EPR correspondence [3,11] in the AdS/CFT context, where entanglement entropy is related to geometric connectivity via the Ryu-Takayanagi formula [4].

3.12. “Instant correlation” → Nonlocal constraint satisfaction (not signal)

The phase consistency principle is a *global* constraint: the inconsistency functional $I[\varphi]$ is evaluated on loops that may span arbitrary distances. When a measurement at one endpoint of an entangled pair projects the local phase structure into a definite state, the global consistency condition determines the correlated outcome at the distant endpoint. No signal propagates between the endpoints; the correlation is *kinematic* (built into the constraint structure) rather than *causal* (transmitted by a propagating disturbance).

This automatically satisfies Bell’s theorem [12] and the no-communication theorem: the correlations violate Bell inequalities (they are genuinely quantum), but they cannot be used to transmit information faster than the emergent light cone permits.

3.13. “Black hole” → Maximally saturated phase knot / causal blockade

LST describes a black hole as a region where the refractive index diverges: $n(r) \rightarrow \infty$ as $r \rightarrow r_s = 2GM/c^2$ (LST Eq. 43 of Ref. [2]). This is the effective-medium description of a horizon. In LPST, the horizon corresponds to the surface where the substrate amplitude vanishes: $A(r) \rightarrow 0$. As $A \rightarrow 0$, the phase stiffness $\kappa = \alpha A^2 \rightarrow 0$, the null cone tips over completely, and outward phase propagation ceases. The “singularity” of general relativity is resolved in LPST by the nonlinear amplitude dynamics: the Mexican-hat potential $V(A)$ in the LPST Lagrangian prevents A from collapsing to a true zero everywhere. The interior of the black hole is a maximally distorted phase configuration — a “phase knot” — in which the substrate retains finite amplitude but the phase gradients are so extreme that no information can propagate outward.

3.14. “Dark matter” → Weakly coupled phase-topology sector

Both LPST and LST require an account of dark matter. In LPST, dark matter candidates are topological defects residing in sectors of the internal polarization manifold M_{int} that couple weakly to the dominant U(1) phase sector responsible for electromagnetism. These defects carry topological charge (and hence mass-energy, via Eq. 7), and they contribute to the correlation kernel $C(x,y)$ (and hence to the emergent metric — they gravitate). But because their polarization structure is orthogonal to the electromagnetic sector, they do not couple to photons: they are gravitationally active but electromagnetically dark. The specific structure of

M_{int} and its homotopy groups determines the dark matter spectrum; this is an open computational problem.

4. Permitted Interpretations: Safe Inter-Theory Statements

When LPST and LST are deployed jointly — as they often will be, given their complementary strengths — certain classes of inter-theory statements are always safe. These are statements that explicitly signal the correct ontological layer, preventing category errors and false contradictions. We define three safety classes.

4.1. Class A — Emergent Language

Statements that explicitly use emergent qualifiers (“behaves as if,” “appears as,” “acts like”) are safe because the qualifier signals that the description is phenomenological, not ontological.

(i) *“Light behaves as if it were the substrate at observable scales.”* — Safe. The qualifier “behaves as if” signals that light’s substrate-like appearance is phenomenological, not ontological. At the LST layer of description, light’s role is indistinguishable from that of a true substrate; the qualifier preserves the distinction without denying the phenomenological accuracy.

(ii) *“Matter appears as trapped light due to phase closure.”* — Safe. The qualifier “appears as” acknowledges that the “trapped light” picture is the LST-level description of what is, at the LPST level, a topologically stabilized phase defect. The causal attribution (“due to phase closure”) correctly identifies the LPST mechanism, grounding the LST intuition in the deeper ontology.

(iii) “*Gravity acts like refraction of the vacuum.*” — Safe. The qualifier “acts like” preserves the distinction between the effective-medium description (LST: gravity is refraction) and the fundamental mechanism (LPST: gravity is phase-consistency-driven metric response). The statement is phenomenologically accurate without being ontologically misleading.

4.2. Class B — Correlation Language

Statements about correlations and constraints that avoid causal language are safe because they do not imply superluminal signaling or causal mechanisms that violate the emergent null-cone structure.

(i) “*Entangled systems behave as a single extended phase object.*” — Safe. This describes the shared topological structure without implying that a signal has been sent between the subsystems. The “single extended phase object” is a statement about the correlation kernel $C(x,y)$, which encodes the non-local phase structure without causal implications.

(ii) “*Correlations are enforced globally by substrate consistency.*” — Safe. This correctly attributes the non-local correlations to the global phase consistency principle (LPST Postulate P3) rather than to a causal mechanism. The consistency principle is a constraint on the space of allowed configurations, not a dynamical process that transmits influence.

4.3. Class C — Effective Limits

Statements about the domain of validity of each description are safe because they explicitly acknowledge the layering structure.

(i) “*At macroscopic scales, LPST reduces to LST phenomenology.*” — Safe. This correctly identifies the coarse-graining relationship: when the microscopic degrees of freedom (amplitude fluctuations, short-wavelength

phase variations) are integrated out, the resulting effective description coincides with LST.

(ii) “*LST captures the dominant low-energy modes of LPST.*” — Safe. This identifies LST as an effective theory valid in a specific energy regime (energies below the scale set by the substrate correlation length ξ), which is the correct characterization.

(iii) “*The LST action is an effective action obtained by integrating out LPST’s amplitude mode.*” — Safe. This states the formal derivation pathway described in Section 2.3, correctly identifying the technical procedure that connects the two theories.

5. Prohibited Phrasings: The Do-Not-Say List

Certain phrasings, if taken literally, create genuine contradictions between the ontological layers or import misleading intuitions that undermine the coherence of the single-substrate program. This section enumerates the canonical prohibited phrasings, explains the problem with each, and provides the corrected form.

5.1. Prohibited: “Light is the substrate”

Problem. This phrasing conflicts with LPST’s phase primacy. In LPST, the substrate is $L = A e^{i\varphi}$ with internal polarization $\mathbf{u}$ — a phase-geometric object whose degrees of freedom include amplitude, phase, and internal polarization. Light (propagating electromagnetic radiation) is the dominant excitation of this substrate, not the substrate itself. Claiming “light is the substrate” conflates the excitation with the ground state. The error is precisely analogous to claiming “sound is air” or “ocean waves are water.” Sound is a propagating disturbance *in* air; ocean waves are propagating

disturbances *of* water. Light is a propagating disturbance *of* the phase substrate. The medium and its excitations are categorically distinct.

Corrected form: *“Light is the dominant propagating excitation of the substrate.”*

5.2. Prohibited: “Entanglement transmits instant physical influence”

Problem. This phrasing violates the Phase-Local Quantum Consistency (PLQC) constraint implicit in LPST’s causal structure. The phase consistency principle is a global constraint, not a causal mechanism. The null cone structure derived from the emergent metric (LPST Section 10 of Ref. [1]) enforces strict causal ordering: no physical influence — no energy, momentum, or information — can propagate outside the local null cone. Entanglement correlations are enforced by the global consistency of the phase configuration, which is a constraint on allowed states, not a dynamical process that transmits influence. Saying “transmits instant physical influence” implies a causal mechanism that violates the emergent spacetime’s causal structure, contradicting both LPST and established physics [12].

Corrected form: *“Entanglement enforces instantaneous correlation without signal propagation.”*

5.3. Prohibited: “Time literally stops inside a black hole”

Problem. This phrasing breaks global update consistency. In LPST, the phase-update ordering defines time (Section 3.7). At a black hole horizon, outward phase propagation halts: the local phase stiffness $\kappa = \alpha A^2$ approaches zero, the null cone closes, and no outward-directed phase perturbation can escape. But this does not mean that the interior phase

dynamics *cease*. The substrate amplitude remains nonzero (bounded below by the Mexican-hat potential), and internal phase dynamics continue — they simply decouple from the exterior. The interior develops its own phase-update ordering, inaccessible to external observers but internally consistent. Saying “time literally stops” confuses the exterior observer’s description (outward signals freeze at the horizon) with the interior dynamics (which continue but are causally disconnected).

Corrected form: *“Outward phase propagation halts at the horizon; internal phase dynamics decouple from the exterior.”*

5.4. Prohibited: “Matter is just a standing wave”

Problem. This phrasing ignores topological protection and is therefore physically incorrect. A standing wave in a linear medium is generically unstable: any sufficiently strong perturbation can disperse it. Derrick’s theorem [7] makes this precise: in three spatial dimensions, static solutions of scalar field theories with only gradient energy and a potential are unstable to rescaling. The theorem is evaded in LPST by the topological stabilizer $K(\varphi, \mathbf{u})$ and the non-trivial homotopy of the target space $S^1 \times M_{\text{int}}$. The topological charge $Q \in \mathbf{Z}$ is exactly conserved; no continuous deformation can change it; and the defect therefore cannot dissipate into the vacuum. The “standing wave” description captures the oscillatory spatial profile of the defect but misses the essential ingredient: the topological obstruction to decay. Without topological protection, the “standing wave” description predicts that all matter should eventually dissipate — which is empirically false.

Corrected form: *“Matter is a topologically stabilized phase structure describable as a standing-wave pattern.”*

5.5. Prohibited: “Gravity is nothing but refraction”

Problem. This phrasing inverts the causal direction. In LST, the refractive index $n(x)$ is treated as the explanatory primitive: gravity is “explained by” refraction of the luminous substrate. In LPST, the causal direction is reversed. The phase-consistency-driven suppression of the substrate amplitude $A(x)$ near topological defects reduces the local phase stiffness $\kappa(x)$, which modifies the correlation kernel $C(x,y)$, which modifies the emergent metric $g_{\mu\nu}$. The refractive index $n(x) = A_0/A(x)$ is a *consequence* of this process, not its cause. Calling gravity “nothing but refraction” mistakes the macroscopic manifestation (refractive behavior) for the microscopic mechanism (phase-consistency-driven metric response). This is analogous to claiming that temperature is “nothing but” the height of mercury in a thermometer: the thermometer reading is a manifestation of temperature, not an explanation of it.

Corrected form: *“Refractive behavior is the macroscopic manifestation of phase-consistency-driven metric response.”*

5.6. Prohibited: “The vacuum is electromagnetic light”

Problem. This phrasing conflates the excitation with the substrate in its ground state. The LPST vacuum is $L_{\text{vac}} = A_0 e^{i\varphi_0}$ — a maximally coherent, zero-gradient phase configuration. It supports electromagnetic modes (propagating φ fluctuations) but is not *composed of* them. The vacuum contains no photons; it is the state from which photons can be excited. This is the standard quantum-field-theoretic understanding (the Fock vacuum is the state annihilated by all annihilation operators), translated into substrate language. Calling the vacuum “electromagnetic light” projects the excitation spectrum back onto the ground state, producing a category error: it attributes to the ground state the properties of its excitations.

Corrected form: “*The vacuum is a balanced, globally coherent phase state supporting electromagnetic modes.*”

6. Cross-Theory Paradox Handling Rules

When LPST and LST are used together, apparent paradoxes may arise from the interaction of their different levels of description. This section formulates three rules for resolving such paradoxes systematically.

6.1. Rule P1 — Layer-Artifact Dissolution

Rule P1

If a paradox disappears when analyzed at the LPST layer, it was an LST artifact — a consequence of the effective description’s limited validity, not a genuine physical contradiction.

Example: The standing-wave stability paradox. In LST, matter is modeled as standing waves of the electromagnetic substrate. But standing waves in linear media are unstable: they dissipate over time due to radiative losses and perturbative instabilities. This creates an apparent paradox within LST: if matter is a standing wave, why doesn’t it dissipate? At the LPST layer, the paradox dissolves immediately. Matter is not merely a standing wave; it is a *topologically stabilized* phase defect carrying an integer-valued topological charge $Q \in \mathbb{Z}$. The charge is exactly conserved by the continuity of the phase and polarization fields. No continuous deformation can reduce Q to zero; the defect cannot decay to the vacuum without encountering an anti-defect of charge $-Q$. The apparent paradox was an artifact of the LST description, which captured the wave-like character of the defect but missed the

topological protection. Derrick’s theorem [7] confirms that the wave pattern alone is unstable; the topological stabilizer $K(\varphi, \mathbf{u})$ is essential.

Other examples of layer-artifact paradoxes include: the apparent violation of energy conservation in expanding spacetime (dissolved by recognizing that LPST’s energy is defined relative to the substrate, not to a background metric); and the apparent paradox of vacuum energy density (dissolved by recognizing that the LPST vacuum has zero phase-gradient energy, and the “vacuum energy” of quantum field theory is an artifact of the effective description’s regularization scheme).

6.2. Rule P2 — Irreducible Paradox Identification

Rule P2

If a paradox survives analysis at the LPST layer — if it cannot be resolved by appealing to the deeper ontology — then it is an irreducible paradox, a genuine closure boundary of the theoretical framework.

Examples. (i) “Why this universe?” — the question of why one particular initial configuration of the phase substrate was realized rather than some other. LPST defines the space of allowed configurations but provides no mechanism for selecting among them. This is an irreducible paradox: it cannot be dissolved by going deeper, because LPST is the deepest layer in the framework. (ii) “Why these initial phases?” — the contingency of the initial conditions $\varphi(x, t_0)$ and $\mathbf{u}(x, t_0)$. No dynamical principle within LPST determines the initial state; the phase consistency principle constrains allowed configurations but does not uniquely select one. These are not failures of the theory; they are the points at which the framework necessarily

contacts the limits of theoretical explanation. Section 7 catalogs all five irreducible paradoxes.

6.3. Rule P3 — Layering Discipline

Rule P3

Never use LST intuition to argue against LPST axioms. LST is not allowed to veto LPST.

This rule guards against a subtle but dangerous error: using the effective description to undermine the microscopic theory from which it is derived. If an LST-derived intuition (e.g., “light should be the fundamental entity because it is the simplest observable”) appears to conflict with an LPST axiom (e.g., “the phase field L , not light, is the fundamental entity”), the LPST axiom takes precedence. LST intuitions are effective-level heuristics: they are valid within their domain but cannot override foundational postulates.

The justification is structural: LST is *derived* from LPST through a coarse-graining procedure (Section 2.3). An effective theory inherits its validity from the microscopic theory but cannot modify the microscopic theory’s axioms, just as the Navier-Stokes equations cannot modify Newton’s laws of motion, and the ideal gas law cannot modify the Boltzmann equation. If an effective-level intuition conflicts with a microscopic axiom, the intuition is wrong — or, more precisely, the intuition is valid only within the effective regime and breaks down when extrapolated beyond it.

7. Shared Irreducible Paradoxes

Both LPST and LST inherit five irreducible paradoxes that constitute the **closure boundaries** of the single-substrate program. These are not flaws,

gaps, or failures. They are the points at which any sufficiently complete theoretical framework necessarily contacts the limits of explanation. We invoke the **Paradox Completion Theorem**: any theory that is sufficiently comprehensive to serve as a candidate theory of everything must contain, at its closure boundary, irreducible paradoxes that the theory itself cannot resolve without invoking external principles. This is the physical analogue of Gödel’s incompleteness theorems in mathematical logic: sufficiently powerful formal systems contain true statements that cannot be proved within the system.

7.1. Paradox 1 — The Singularity of Realization

Why does exactly one configuration of the substrate exist, rather than the space of all possible configurations?

LPST defines the space of all physically allowed configurations — the space of all smooth sections $L: X \rightarrow \mathbf{C} \times M_{\text{int}}$ satisfying the phase consistency condition. But exactly one configuration is realized: the actual universe. The substrate’s initial configuration is *contingent* — it could have been different — yet the theory provides no mechanism for selecting among the possibilities. This is not a problem unique to the single-substrate program; it confronts every physical theory. But within the present framework, it is explicitly identified as a closure boundary: the question “why this configuration?” cannot be answered from within the theory.

7.2. Paradox 2 — The Origin of Ordering

What initiated the first phase update?

Time in LPST is the ordering imposed by the phase-update structure of the substrate (Section 3.7). But the first update — the first tick of the substrate’s internal clock — cannot be explained by prior updates, because there are no prior updates. The first update is either uncaused (violating the explanatory

structure of the theory) or caused by something outside the substrate (violating ontological monism). This is the substrate-theoretic analogue of the cosmological initial-singularity problem and the question “what happened before the Big Bang?” It is irreducible within the single-substrate framework.

7.3. Paradox 3 — Observer Self-Reference

How does a subsystem of the substrate (an observer, itself a topological defect) develop a complete description of the substrate?

An observer in LPST is a complex topological defect — a structured configuration of phase and polarization fields — embedded in the very substrate it seeks to describe. The observer is a part attempting to model the whole. This creates a physical instantiation of Gödel-type self-reference limitations: the observer’s model of the substrate is necessarily a subsystem of the substrate, and therefore cannot contain more information than the substrate itself. A complete description of the substrate would require a model at least as complex as the substrate, which cannot fit inside a proper subsystem. The observer’s description is necessarily incomplete, approximate, and self-referentially limited.

7.4. Paradox 4 — Existence Without Strong Interaction

Can the substrate contain sectors that exist (carry phase structure) but never interact with the observable sector?

LPST’s internal polarization manifold M_{int} may have multiple topological sectors — connected components of the defect classification space — some of which couple to the electromagnetic (phase-gradient) sector and some of which do not. Defects in weakly coupled sectors would gravitate (they modify the correlation kernel and hence the metric) but would be electromagnetically invisible: they would constitute dark matter or, more

speculatively, entirely decoupled sectors. The paradox is: if a sector is entirely decoupled from all observable sectors — if it neither gravitates, radiates, nor interacts in any way — does it “exist” in any physically meaningful sense? The substrate formalism assigns it phase structure and topological charge, but no observation, even in principle, could confirm its presence. This is the single-substrate version of the problem of unobservable ontology.

7.5. Paradox 5 — Continuum Giving Rise to Discreteness

How does a fundamentally continuous field give rise to exactly discrete quantum numbers?

The substrate field L is a smooth section of a fiber bundle — a fundamentally continuous object. Yet the physical observables it generates include exactly discrete quantum numbers: integer topological charges, quantized angular momenta, discrete particle masses. LPST’s answer is topological quantization: the integers arise from the homotopy groups of the target space ($\pi_1(S^1) \cong \mathbf{Z}$, $\pi_2(S^2) \cong \mathbf{Z}$, $\pi_3(S^2) \cong \mathbf{Z}$). This explains *how* discreteness arises from continuity, but it does not explain *why* the target-space topology has the particular homotopy groups it does. Why is M_{int} a sphere rather than a torus? Why does $\pi_3(S^2) \cong \mathbf{Z}$ rather than some other group? These questions about the topology of the target space are not answerable from within the theory; they are boundary conditions on the framework itself.

Status. These five paradoxes are not flaws of the single-substrate program. They are its *closure boundaries*, mandated by the Paradox Completion Theorem. Any attempt to resolve one of these paradoxes from within the framework would require extending the framework — adding new postulates, new substrate sectors, or new meta-theoretical principles — which would merely relocate the closure boundary rather than eliminate it.

8. Forbidden Claims: What Each Theory Is Not Allowed to Assert

To maintain the integrity of the compatibility layer, each theory is explicitly prohibited from making certain claims. These prohibitions are not arbitrary; they follow from the structural position of each theory within the layered framework.

8.1. LPST Is Forbidden from Claiming:

(i) “Light is fundamental.” Light — electromagnetic radiation, propagating phase perturbations — is a derived excitation of the substrate, not a postulated primitive. LPST’s four postulates refer to the complex field $L = A e^{i\varphi}$ with internal polarization $\mathbf{u}$, not to electromagnetic radiation. Light emerges as the dominant propagating mode of the phase field in the vacuum sector; it is a consequence of the postulates, not one of them. Claiming that light is fundamental within LPST would conflate the excitation with the substrate and collapse the layering that distinguishes LPST from LST.

(ii) “All phenomenology is obvious from phase geometry.” While LPST aspires to derive all physics from its four postulates, the derivation is far from complete. The effective phenomenology — the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$, the three generations of fermions, the specific values of particle masses and coupling constants, the cosmological constant — requires extensive computation and has not yet been fully derived from the postulates. The claim that it is “obvious” would be premature, scientifically irresponsible, and misleading. LPST claims that the phenomenology is *in principle derivable*; it does not claim that the derivation is easy or that all results are currently known.

(iii) “Paradoxes are eliminated.” LPST dissolves many LST-layer paradoxes (Section 6, Rule P1) by providing the deeper ontological analysis

that renders them artifacts of the effective description. But LPST inherits five irreducible paradoxes (Section 7) that constitute the closure boundaries of the single-substrate program. A claim of total paradox elimination would be false and would violate the Paradox Completion Theorem. The correct claim is: “LPST dissolves effective-layer paradoxes but inherits irreducible closure-boundary paradoxes.”

8.2. LST Is Forbidden from Claiming:

(i) “Instant causation.” Entanglement correlations in LST are structural constraints inherited from the phase consistency principle of LPST. They are not causal influences. Any claim of instant causation — that a measurement at one location *causes* an effect at a distant location instantaneously — violates the null-cone causal structure derived from the emergent metric. The emergent spacetime of LPST has a well-defined causal structure (LPST Section 10 of Ref. [1]): no energy, momentum, or information can propagate outside the local null cone. Correlations can be instantaneous (they are kinematic, built into the constraint structure); causation cannot. LST must use correlation language, not causal language, when describing entanglement phenomena.

(ii) “Substrate propagation literally halts.” At a black hole horizon, *outward* propagation halts: the effective speed of phase perturbations relative to distant observers approaches zero as the horizon is approached. But the substrate does not freeze in absolute terms. The amplitude A approaches zero near the horizon, causing the local phase stiffness to vanish and the null cone to close; but inside the horizon, the substrate retains nonzero amplitude (bounded by the Mexican-hat potential) and continues to evolve. Internal phase dynamics decouple from the exterior but do not cease. Claiming that “substrate propagation literally halts” confuses the exterior observer’s description with the interior reality.

(iii) “Standing waves explain stability by themselves.” As established in Section 5.4, standing-wave patterns in three spatial dimensions are generically unstable by Derrick’s theorem [7]. Stability in LPST requires topological protection: the integer-valued topological charges carried by defects are exactly conserved and prevent decay to the vacuum. The “standing wave” description captures the spatial profile of the defect but does not explain its persistence. LST must acknowledge topological stabilization whenever discussing the stability of matter.

(iv) “Refractive index is fundamental.” The refractive index $n(x) = A_0/A(x)$ is a derived ratio of vacuum to local phase stiffness. It has no independent equation of motion; its value at every point is determined by the amplitude field $A(x)$, which is itself determined by the global phase consistency condition and the topological defect content. Treating $n(x)$ as a fundamental field would incorrectly double-count degrees of freedom and obscure the derivation pathway from LPST to LST.

9. Formal Compatibility Statement

9.1. The Canonical Statement

The formal compatibility relation between LPST and LST is stated as follows:

LPST defines the space of physically allowed configurations through its four postulates and the phase consistency principle. LST describes the effective phenomenology that emerges when the phase dynamics of the substrate are dominated by the propagating modes — the luminous sector — in the low-energy, long-wavelength regime.

More concisely:

*LST is true as an effective description; LPST is true as an ontological one.
Any apparent contradiction arises from mistaking one layer for the other.*

9.2. Domain of Validity

The domain of validity of each theory is precisely characterized by the following conditions.

LST is valid wherever all of the following conditions hold simultaneously:

1. The amplitude mode is frozen: $A(x) \approx A_0$, with fluctuations $\delta A/A_0 \ll 1$.
2. The characteristic wavelength λ of the relevant dynamics is much larger than the substrate correlation length: $\lambda \gg \xi$.
3. The phase and polarization fields dominate the dynamics: the energy density is dominated by $(\nabla\phi)^2$ and $(\nabla\mathbf{u})^2$ terms rather than amplitude gradients $(\nabla A)^2$.
4. The emergent metric $g_{\mu\nu}$ is smooth: curvature invariants $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ are small compared to ξ^{-4} .

These conditions are satisfied in all experimentally accessible regimes: laboratory physics, astrophysics, and cosmology at redshifts $z \ll z_{\text{Planck}}$.

LPST is valid everywhere, including regimes where LST breaks down:

5. Planck-scale energy densities, where the amplitude mode is dynamically active and A deviates significantly from A_0 .
6. Black hole interiors, where $A \rightarrow 0$ and the effective-medium description (refractive index) diverges.

7. The very early universe (the first phase-update cycles), where the substrate may not yet have reached the low-energy regime in which LST's coarse-graining is valid.
8. Topological phase transitions, where the defect content of the substrate changes discontinuously and the smooth-metric assumption fails.

9.3. The One-Sentence Summary

The entire content of this compatibility specification can be distilled into a single canonical statement:

LPST is the hardware that enforces possibility; LST is the software that explains appearance. Conflicts only arise when software metaphors are mistaken for hardware truths.

This statement encodes the layering principle (Section 2), the concept-mapping dictionary (Section 3), the safety classes (Section 4), the prohibited phrasings (Section 5), and the paradox-handling rules (Section 6) in a form suitable for rapid deployment. When in doubt about any inter-theory statement, the practitioner should ask: "Am I describing hardware or software?" If the statement describes software (effective phenomenology, macroscopic observables, intuitive metaphors), it belongs to LST and must be qualified with emergent language. If the statement describes hardware (phase configurations, topological charges, consistency constraints), it belongs to LPST and is ontologically binding.

10. Discussion and Outlook

The compatibility layer established in this paper serves a practical purpose beyond formal clarity: it provides a shared language and a set of navigational rules for the ongoing development of the single-substrate program. As LPST and LST mature, researchers will need to move fluently between the two levels of description, using LST’s intuitive language for phenomenological calculations and LPST’s rigorous formalism for foundational arguments. The compatibility layer ensures that this movement can occur without ontological confusion.

Several open problems in the single-substrate program require coordinated LPST-LST analysis:

(a) Derivation of the Standard Model gauge group from M_{int} . The gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ of the Standard Model must emerge from the topology and geometry of the internal polarization manifold M_{int} . This is a problem that must be addressed at the LPST level (it concerns the fundamental topology of the substrate) but whose solution will be expressed in LST language (gauge bosons as specific defect-sector excitations, coupling constants as topological invariants). The concept-mapping dictionary (Section 3) will be essential for translating between the two descriptions.

(b) Calculation of particle mass spectra from topological defect energetics. The masses of elementary particles must be computed from the energy functional (Eq. 7) evaluated on the topological defect solutions. This requires solving nonlinear partial differential equations on the substrate — an LPST-level calculation — but the results will be compared with the mass formulas of LST (e.g., $m = \hbar/(rc)$) to determine the effective geometric parameters (toroidal radius, winding number) of each particle species. The compatibility layer’s concept mapping ensures that the comparison is made

correctly, with LPST masses treated as fundamental and LST mass formulas treated as effective approximations.

(c) Cosmological implications. Inflation, dark energy, and the large-scale structure of the universe must be derived from substrate dynamics. The early universe — the epoch of the first phase-update cycles — lies outside the domain of validity of LST (Section 9.2) and must be described entirely in LPST language. As the universe cools and the amplitude mode freezes, LST becomes applicable, and the transition between the two descriptions must be handled using the layering principle (Section 2). The compatibility layer provides the rules for this transition. Similarly, dark energy may be interpretable as a residual phase inconsistency at cosmological scales — a nonzero value of $\mathbb{I}[\phi]$ on the largest contractible loops — which in LST language appears as a cosmological constant.

(d) Experimental signatures distinguishing LPST/LST from GR+QFT.

The single-substrate program makes predictions that differ from general relativity plus quantum field theory, particularly in extreme regimes. These include: modifications to the dispersion relation at Planck-scale energies (detectable, in principle, through gamma-ray burst timing [5]); corrections to black hole thermodynamics arising from the Mexican-hat potential’s prevention of true $A = 0$ singularities; and signatures in the cosmic microwave background from topological phase transitions in the early substrate. The compatibility layer will be essential for translating LPST-level predictions into observationally accessible LST-level signatures.

We note that the compatibility layer is itself a **living document**. As LPST and LST develop, new concept mappings will need to be added to the dictionary (Section 3), new prohibited phrasings may be identified (Section 5), and new paradox-dissolution examples may clarify the boundary between layer-artifact and irreducible paradoxes (Section 6). The five irreducible paradoxes (Section 7) may also evolve: a future extension of the framework might

resolve one of them, revealing it to have been a layer artifact after all, while simultaneously exposing a new closure-boundary paradox. The Paradox Completion Theorem guarantees that the total count of irreducible paradoxes can never reach zero within a sufficiently comprehensive framework.

Finally, we observe that the relationship between LPST and LST may itself serve as a paradigm for inter-theory relationships in physics more broadly. The layering principle, the concept-mapping dictionary, the safety classes, the prohibited phrasings, and the paradox-handling rules are formal tools that could, in principle, be applied to any pair of theories related by an effective-field-theory derivation: quantum chromodynamics and chiral perturbation theory, the Standard Model and Fermi theory, string theory and supergravity, or any future pair of theories occupying different levels of ontological depth.

11. Conclusion

We have established a complete formal compatibility layer between Light-Phase Substrate Theory (LPST) and Luminous Substrate Theory (LST). The compatibility layer consists of nine components:

9. The **Layering Principle** (Section 2): LPST occupies the ontological hardware layer; LST occupies the effective phenomenological layer. The derivation is strict and unidirectional.
10. The **Non-Negotiable Layering Rule** (Rule 1): any LST statement that appears fundamental must be reinterpreted as emergent from LPST.
11. The **Concept Mapping Dictionary** (Section 3): a complete translation table from LST constructs to LPST antecedents, covering fourteen fundamental concepts.

12. The **Variable Status Rule** (Rule 2): LST variables are coarse-grained observables, not primitives.
13. The **Three Safety Classes** (Section 4): emergent language, correlation language, and effective-limit statements are always permitted.
14. The **Prohibited Phrasings** (Section 5): six canonical do-not-say phrases with their corrected forms.
15. The **Paradox Handling Rules** (Section 6): three rules for resolving apparent inter-theory paradoxes.
16. The **Irreducible Paradoxes** (Section 7): five closure-boundary paradoxes that the framework cannot resolve internally.
17. The **Forbidden Claims** (Section 8): specific assertions that each theory is prohibited from making.

The central principle governing the entire compatibility layer is the strict, unidirectional layering of the two theories: *LPST defines what reality is allowed to be; LST describes how that reality appears when phase dynamics are dominated by luminous propagation*. Every rule, mapping, prohibition, and paradox classification in this document follows from this single principle.

We close with the canonical summary statement:

LPST is the hardware that enforces possibility; LST is the software that explains appearance. Conflicts only arise when software metaphors are mistaken for hardware truths.

References

- [1] M. Hanks, “Light-Phase Substrate Theory (LPST): A Relational Phase-Interference Model of Emergent Space, Time, Matter, and Gravity,” arXiv:2601.XXXXX [hep-th] (2026).

- [2] M. Hanks, "The Luminous Substrate: A Unified Ontology of Emergent Spacetime and Matter," Preprint (2026).
- [3] J. Maldacena, "The large- N limit of superconformal field theories and supergravity," *Adv. Theor. Math. Phys.* **2**, 231 (1998).
- [4] S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy from AdS/CFT," *Phys. Rev. Lett.* **96**, 181602 (2006).
- [5] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260 (1995).
- [6] T. H. R. Skyrme, "A unified field theory of mesons and baryons," *Nucl. Phys.* **31**, 556 (1962).
- [7] G. H. Derrick, "Comments on nonlinear wave equations as models for elementary particles," *J. Math. Phys.* **5**, 1252 (1964).
- [8] W. Gordon, "Zur Lichtfortpflanzung nach der Relativitätstheorie," *Ann. Phys.* **377**, 421 (1923).
- [9] J. A. Wheeler, "Geons," *Phys. Rev.* **97**, 511 (1955).
- [10] E. Verlinde, "On the origin of gravity and the laws of Newton," *JHEP* **2011**, 029 (2011).
- [11] M. Van Raamsdonk, "Building up spacetime with quantum entanglement," *Gen. Rel. Grav.* **42**, 2323 (2010).
- [12] J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics* **1**, 195 (1964).